

# **PROJECT PROPOSAL**

## **PREDICTING EPL MATCH OUTCOMES USING STATISTICAL LEARNING**

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## 1. Description of the Problem

Sports analytics has transformed from a niche interest into a multibillion dollar industry used by professional clubs and betting markets. However, accurately predicting the outcome of a football (soccer) match is notoriously difficult due to the low-scoring nature of the game and the high degree of randomness.

The goal of this project is to develop a statistically sound machine learning framework to predict the outcome of English Premier League (EPL) matches (Home Win, Draw, Away Win). Rather than focusing solely on classification accuracy, this project aims to estimate well-calibrated probabilities. This approach allows for a deeper understanding of the stochastic processes driving match results, such as team attack strength and defense strength.

## 2. Key Questions to Answer

- **Feature Importance:** Which in-game statistics (e.g., Shots on Target, Corners, or recent form) are the most reliable predictors of future match outcomes?
- **Home Advantage:** How can we quantify the statistical significance of playing at home, and does it vary between teams?
- **Model Calibration:** Can a statistical model produce probability estimates that are consistent with observed frequencies, effectively "beating" naive baselines?

## 3. Dataset Description

The project will use the [Football-Data.co.uk](https://www.football-data.co.uk) repository, specifically the English Premier League match data covering the last 5 completed seasons. This dataset is publicly available and machine-readable. It includes the following key features:

### Match Outcomes

- FTR: Full Time Result (H, D, A)
- FTHG / FTAG: Full Time Home Goals and Away Goals

### Predictive Signals (Performance Metrics)

- HS / AS: Home Team Shots / Away Team Shots
- HST / AST: Home/Away Shots on Target
- HC / AC: Home/Away Corners
- HY / AY: Home/Away Yellow Cards

### Betting Odds (Benchmark)

- B365H, B365D, B365A: Odds from Bet365, serving as an information-efficient baseline for model evaluation for games yet to happen.

## 4. Expected Outcome

- A supervised predictive model that outputs probabilities for match outcomes.
- An analysis of feature coefficients to interpret which tactical metrics (e.g., defensive solidity vs. attacking volume) drive success.
- A rigorous evaluation using to demonstrate the model's theoretical validity compared to baseline heuristics.

## 5. Challenges

- Football matches are highly stochastic; the "better" team does not always win. This sets a low theoretical ceiling on accuracy (typically ~55-60%).
- Team performance metrics (e.g., Shots and Goals) are highly correlated, requiring regularization (Lasso/Ridge) to prevent overfitting.
- Team strength changes over time (transfers, injuries), meaning a model trained on data from 5 years ago may not apply to today without temporal weighting or rolling-window validation.